

Additional file 2

Case-level audit of thyroid ultrasound and cytology deep learning: label conflicts, split design, and shortcut-sensitive performance in a public archive

Supplementary Figures S1–S12. Vector (PDF/SVG) versions of all statistical figures are in the analysis release.

Fig. S1 Fold-level AUROC under grouped five-fold cross-validation for each backbone and the ensemble; cytology (left) and ultrasound (right). Fold values are shown for completeness; the primary estimate is the pooled out-of-fold AUROC (Fig. 2; Table 2; Supplementary Table S6).

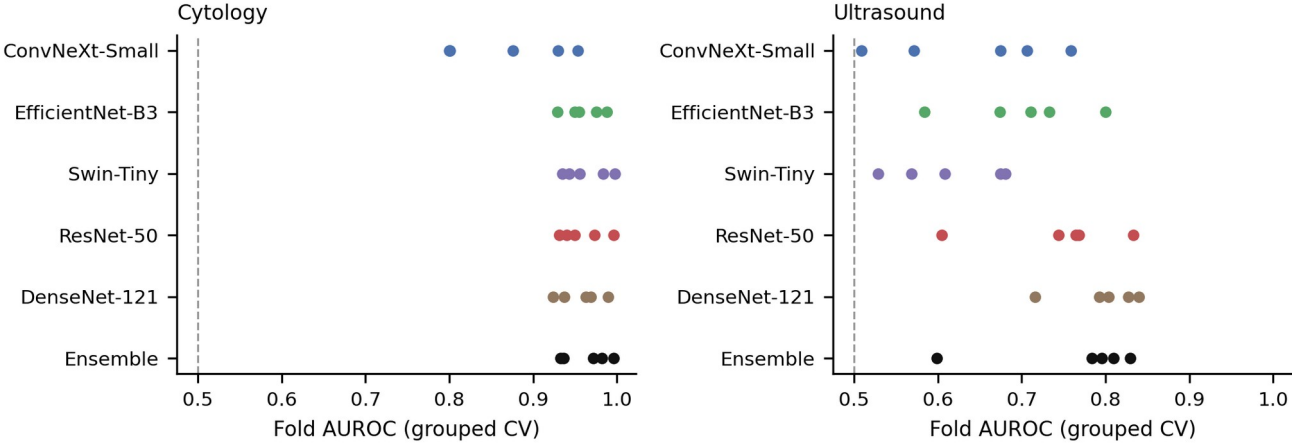


Fig. S2 Masking QA: random sample (fixed seed) of 12 ultrasound frames in which a sonographic sector was detected (solid box).

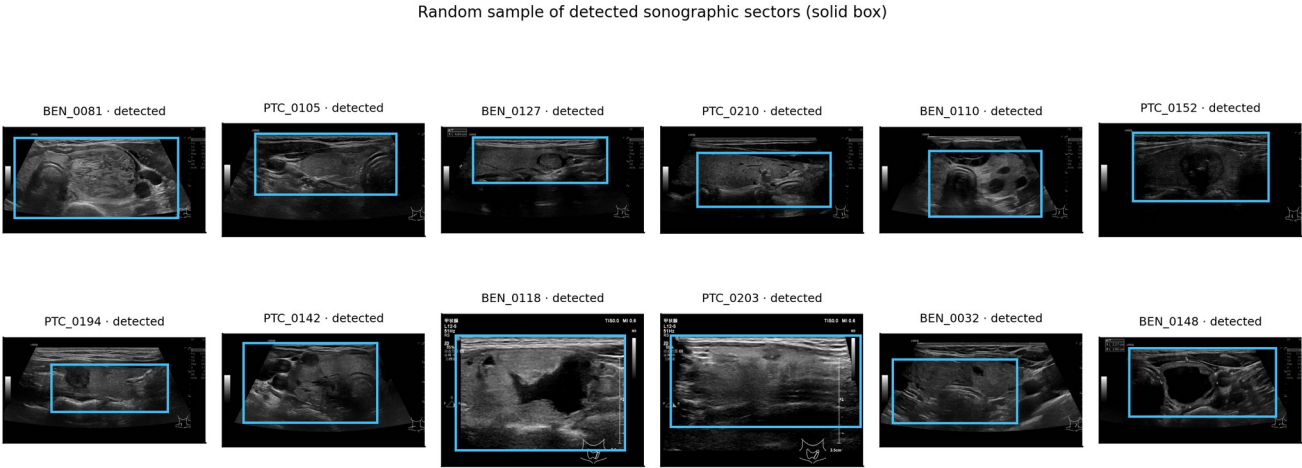


Fig. S3 Masking QA: random sample (fixed seed) of 12 frames in which detection failed and the fixed central box was used (dashed box). In all 88 fallback frames the largest textured region covered less than 25% of the frame (Supplementary Table S9).

Random sample of fallback boxes (dashed): detection failed, fixed central box used

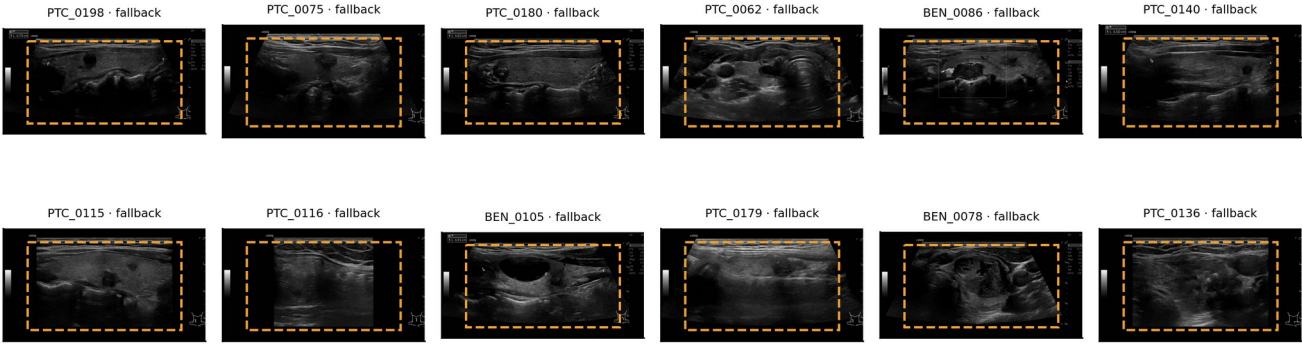


Fig. S4 Masking QA: the six detected boxes with the smallest areas, the candidates most likely to cover the sonographic sector only partially.

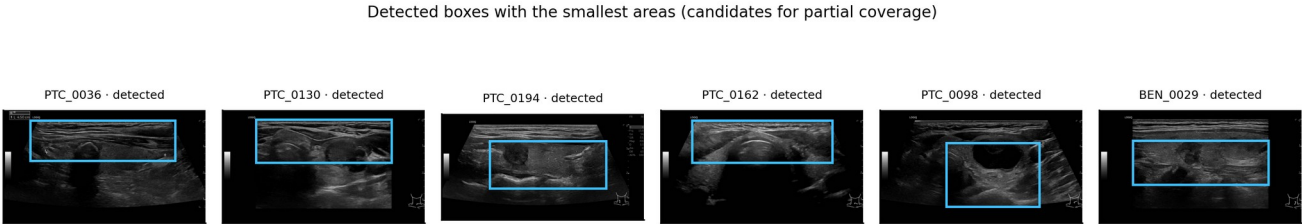


Fig. S5 Grad-CAM (exploratory) for the frozen ultrasound EfficientNet-B3: one representative test-set case per outcome (median predicted probability within TP, TN, FP and FN). Raw attribution is shown on one shared scale (0 to the 99th percentile of positive attribution across all panels) with the perceptually uniform magma colour map. Solid box: heuristically detected sonographic region ("detected region"); dashed box: fallback box. Attribution maps do not establish the anatomical source of predictions.

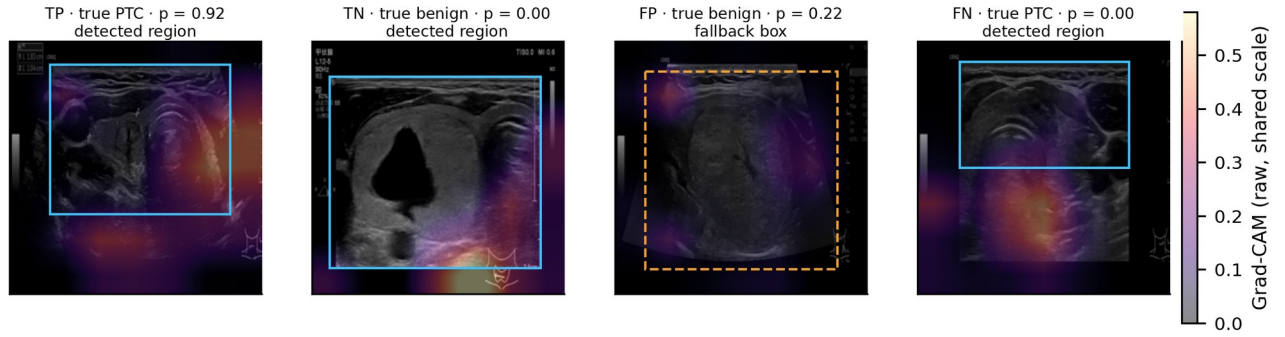


Fig. S6 As Fig. S5, extended to the lowest and highest predicted probability within each outcome, on the same shared scale.

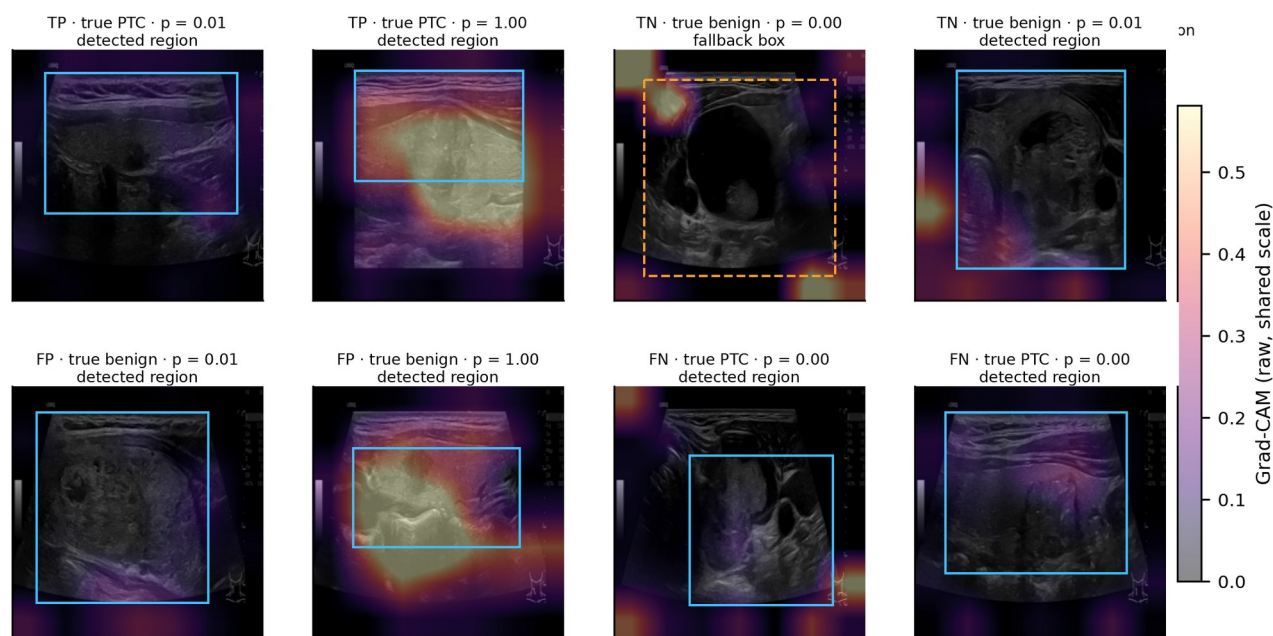


Fig. S7 Exploratory two-dimensional visualisation (t-SNE) of penultimate-layer embeddings of the frozen ultrasound EfficientNet-B3 for the development archive, the TN3K test set and DDTI. Cluster spacing is not a measure of distance between cohorts; quantitative comparison uses maximum mean discrepancy (Supplementary Table S15).

Exploratory two-dimensional visualization (t-SNE);
cluster spacing is not a measure of distance between cohorts

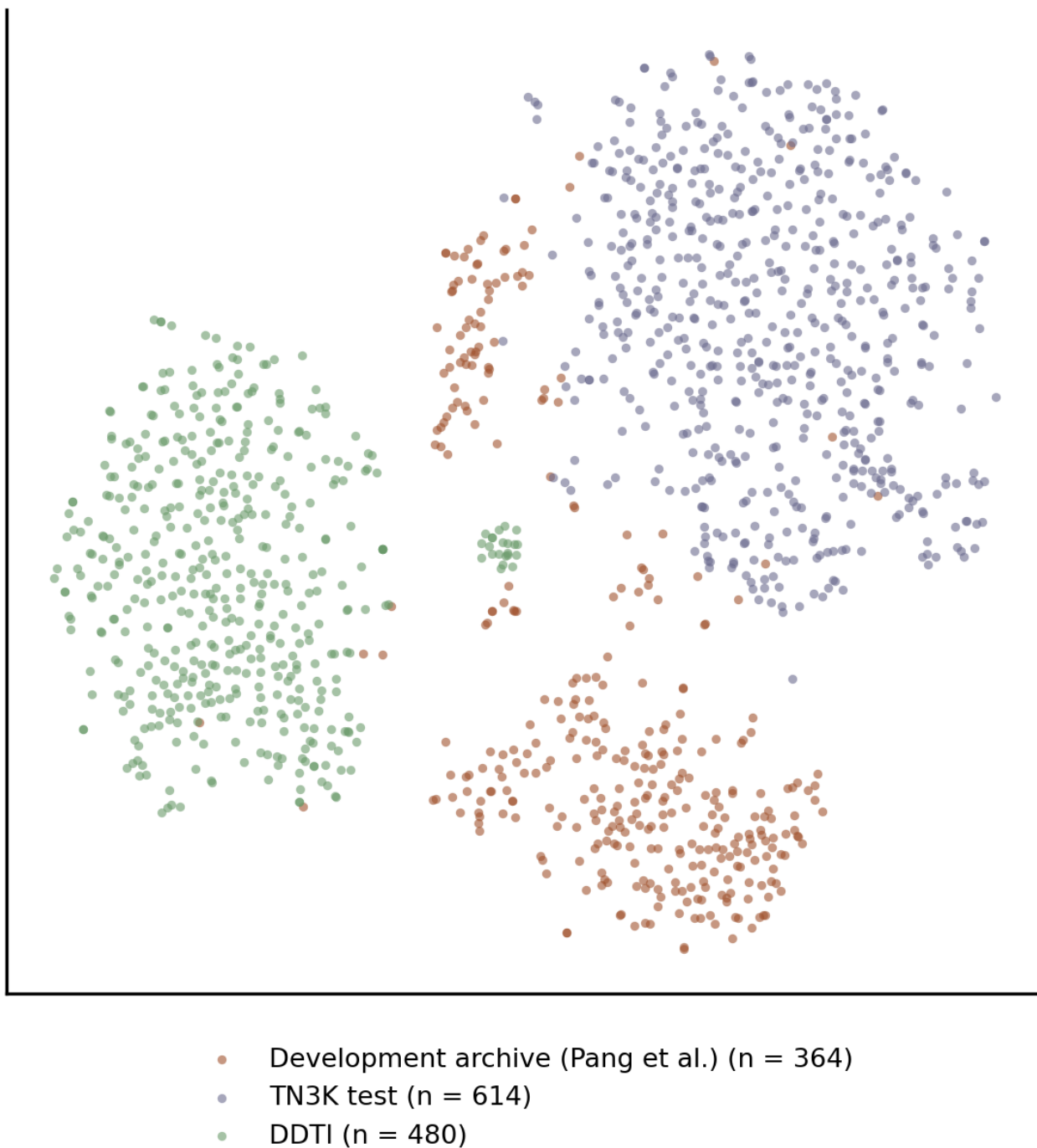


Fig. S8 Post-exposure external reassessment, all backbones and the ensemble: TN3K test set against distributed benign/malignant labels (left) and DDTI concordance with the TI-RADS 2–3 versus 4a–5 grouping, a suspicion category rather than malignancy (right).

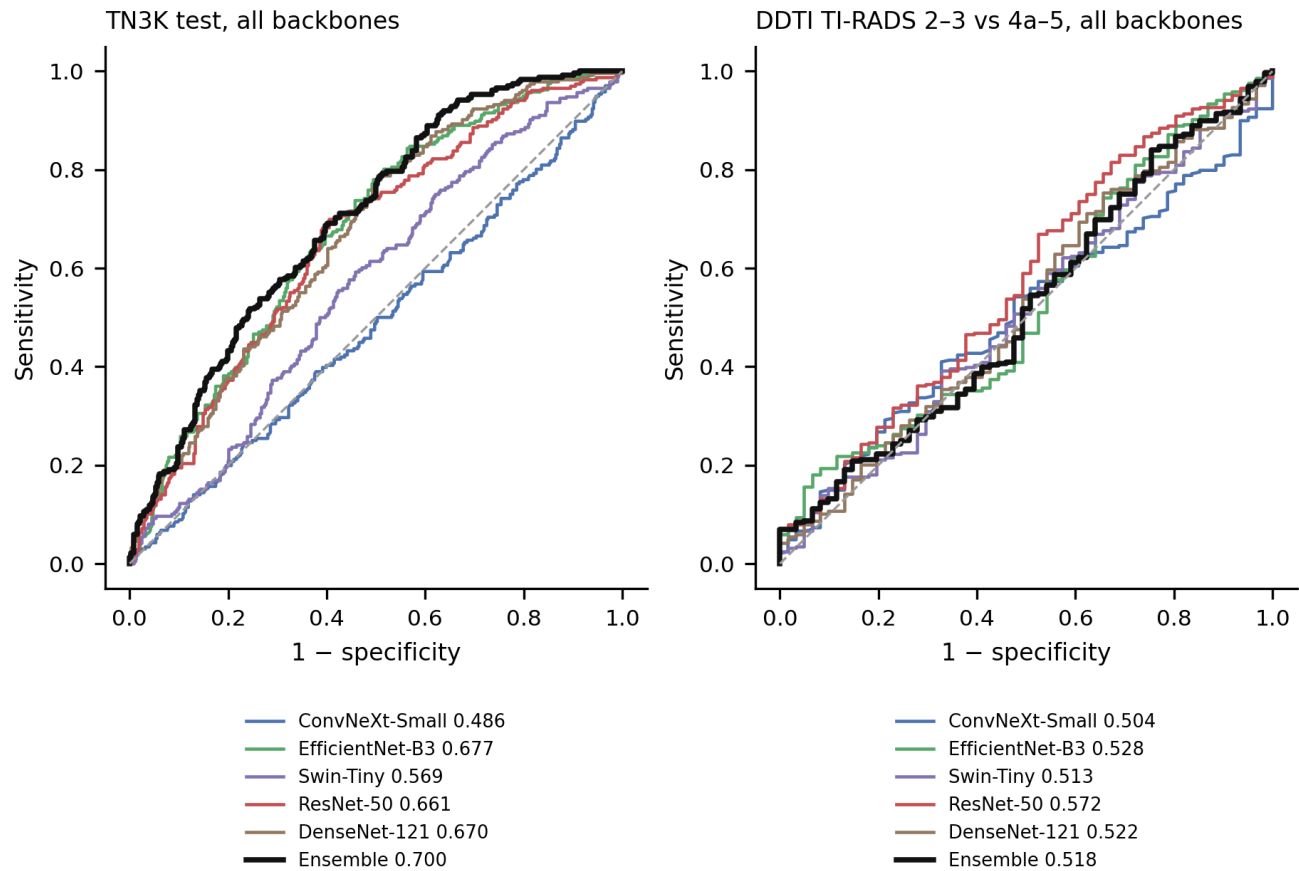


Fig. S9 Internal calibration of the ensembles on the frozen test sets, shown separately by modality, with five equal-count bins and the number of images per bin.

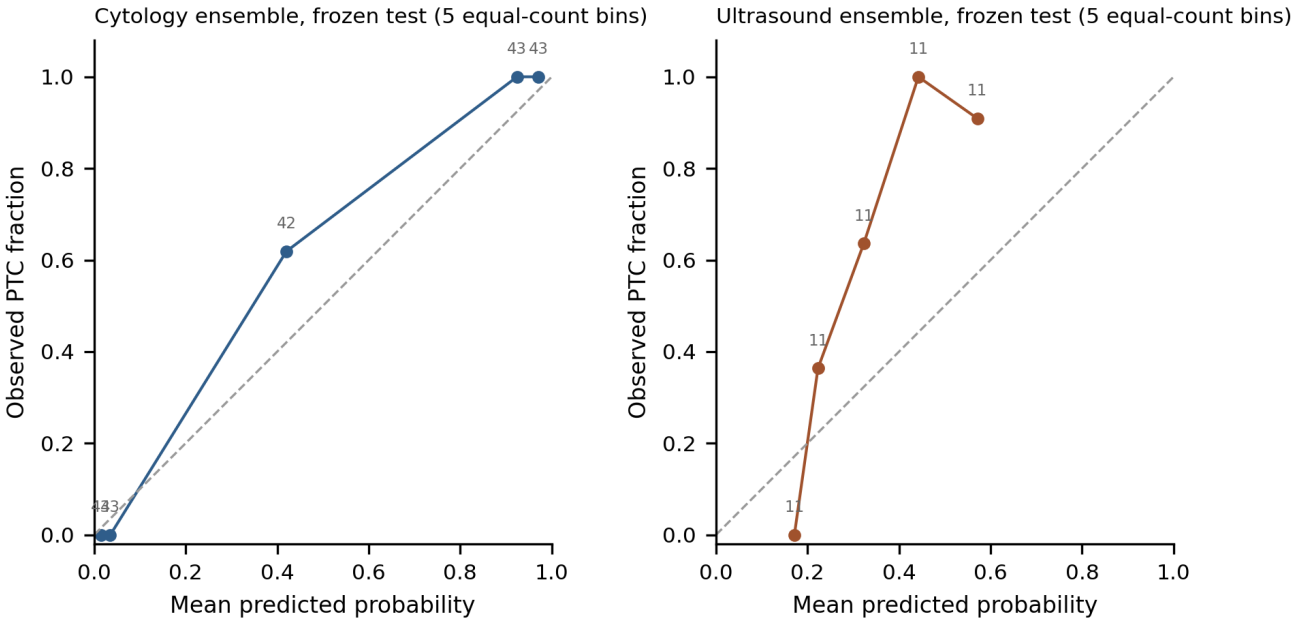


Fig. S10 Cytology frozen test set at block (image) level, ensemble. The primary analysis is at case level (Fig. 2C).

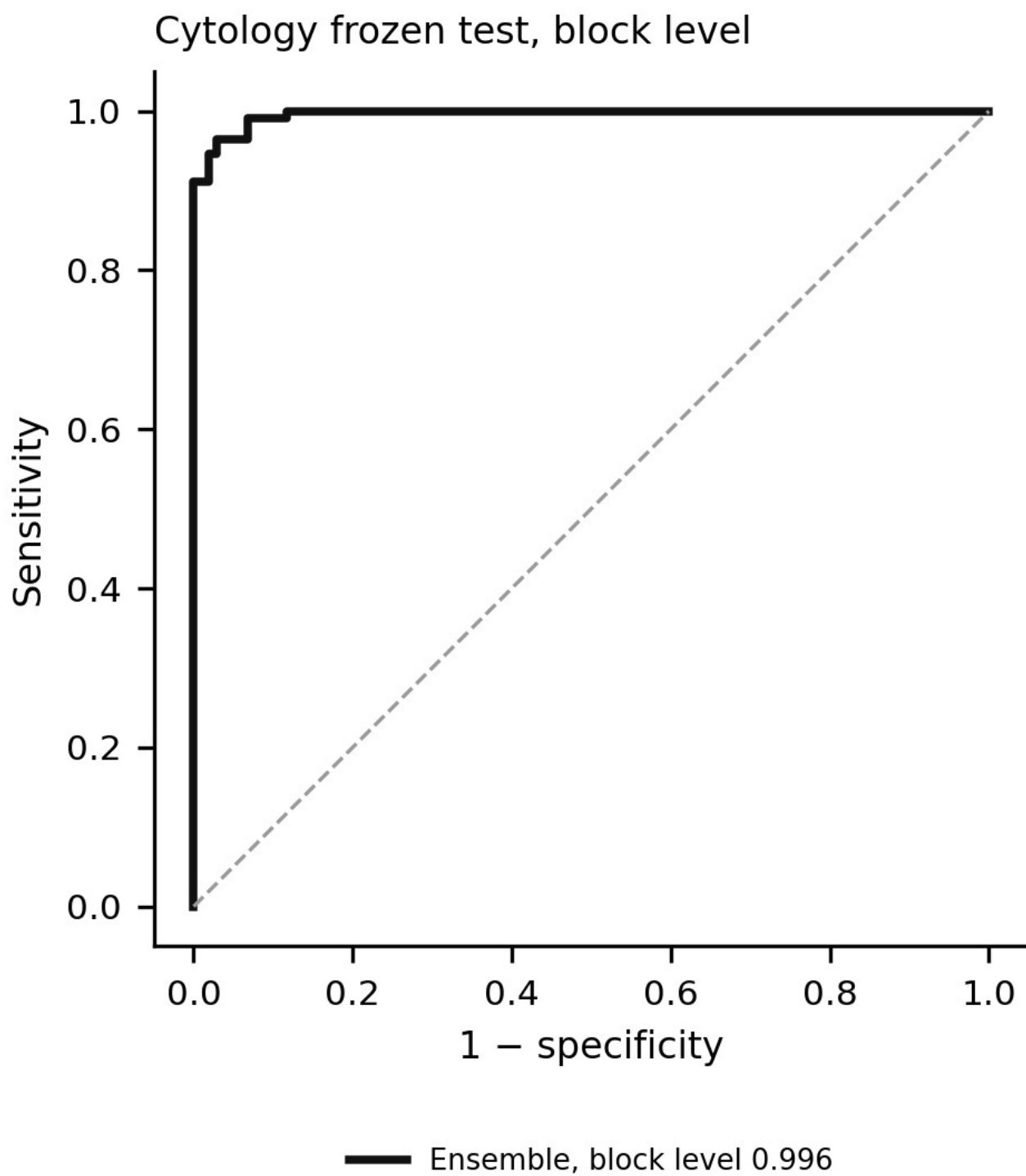


Fig. S11 Near-duplicate adjudication: every perceptual-hash pair at Hamming distance ≤ 6 whose members fall in different frozen partitions, with partition, Hamming distance, SSIM and verdict (near-duplicate if $\text{SSIM} \geq 0.90$; Supplementary Table S3).

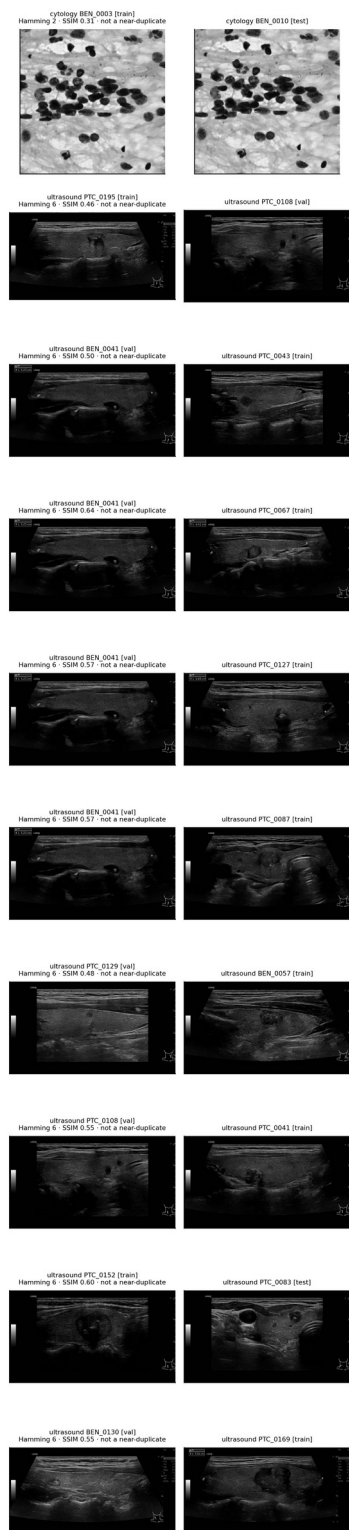


Fig. S12 Acquisition context: the four most frequent ultrasound frame sizes, with class counts, and one randomly drawn image per class from each (fixed seed).

